Individual Progress Report

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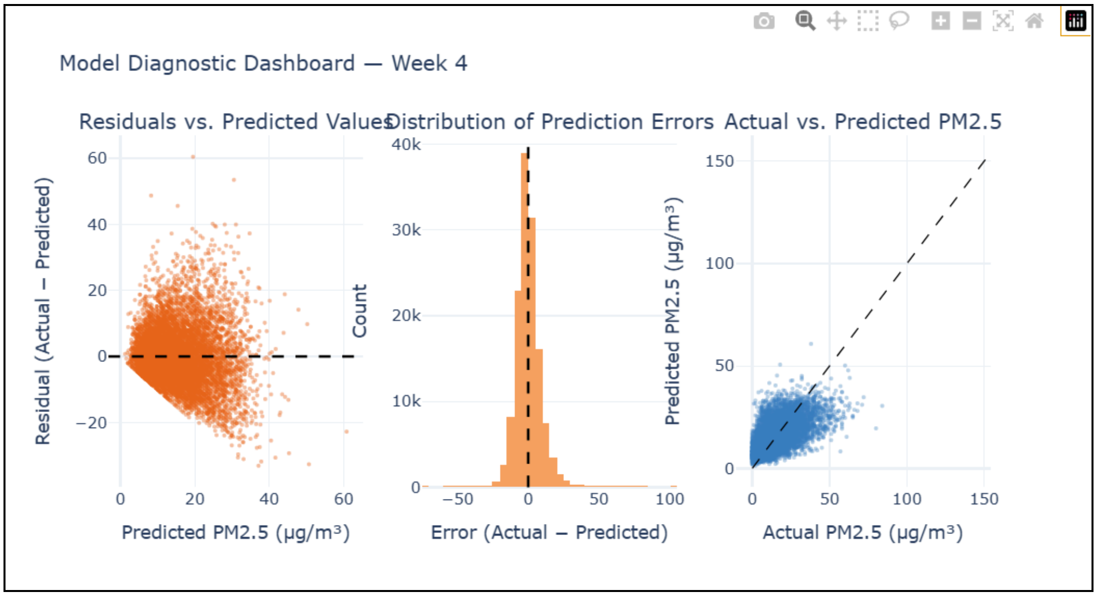
Group Topic: **Urban Air Quality & Environmental Health**	Lab No. Week 4
Git Repo/ Colab Link: https://github.com/Cristy16/BDA---Final-Project	Date: 05/16/26

Tableau Link

Group Objective: To transition the Urban Air Quality prediction model into a production-ready pipeline, integrate external tools, and extend the analysis with unsupervised pattern mining. This week also required visualizing processed data in a professional format as part of the lab requirement.

Tasks Completed

1. Diagnostic Visualizations section (Section 3) of the team Jupyter Notebook, including the Markdown documentation explaining the purpose of each chart and the Python code cell generating the three-panel interactive dashboard using Plotly.
2. Computed the residuals from the pipeline's test set predictions (`residuals = y_test.values - y_pred`) on the 134,972-row test set and downsampled to 10,000 points for rendering performance.
3. Built a three-panel interactive diagnostic dashboard (`fig_diag`) using Plotly's `make_subplots`:
   - **Panel 1 — Residuals vs. Predicted Values:** Scatter plot of prediction errors against predicted PM2.5 values, with a zero-reference dashed line. Used to detect heteroscedasticity (funnel shapes indicating unequal error variance).
   - **Panel 2 — Distribution of Prediction Errors:** Histogram of all residuals with 60 bins and a zero-reference line. A bell curve centered at zero indicates random, non-systematic errors.
   - **Panel 3 — Actual vs. Predicted PM2.5:** Scatter plot of true vs. predicted values against the perfect-prediction diagonal line. Points close to the diagonal represent accurate predictions.



4. Applied interactive features (hover tooltips, zoom, pan) via Plotly, satisfying the lab's *professional format* visualization requirement — an upgrade from the static matplotlib charts used in Week 3.

Tasks to be Completed

I was able to complete all task assignments given to me this Week 4 activity, which is the Diagnostic Visualizations section. The submission of the finalized notebook to the team Git repository was handled by Polly, the team lead and member assigned to that task.

Results and Analysis

The diagnostic visualizations were rendered successfully on the Week 4 pipeline's test set output. The pipeline itself produced the following metrics, which provide context for interpreting the charts:

Metric	**Value**
MAE	6.2320 μg/m³
RMSE	8.4123 μg/m³
R^2 (Test Set)	0.3622
R^2 (5-Fold CV Mean)	0.3563

CV Std	0.0023

Note on R^2 vs. Week 3: The R^2 this week (0.3622) is lower than Week 3's 0.5237. This is expected — the pipeline this week uses a different data split from the full `clean_air_quality_data.csv` (674,856 rows after cleaning), applied to a rebuilt pipeline with StandardScaler prepended. The cross-validation mean of 0.3563 with a standard deviation of only 0.0023 confirms the model is stable and not overfitting.

Interpretation of the three diagnostic charts:

- **Residuals vs. Predicted:** A random scatter around zero with no clear funnel or curve pattern confirms the model errors are not systematically biased. However, spread increases at higher predicted PM2.5 values, consistent with the extreme pollution spike events identified in earlier phases that the model has difficulty capturing.
- **Error Distribution:** The histogram approximates a bell curve centered near zero, indicating errors are largely random rather than directional. The slight right tail corresponds to cases where the model underpredicts severe pollution spikes.
- **Actual vs. Predicted:** Points cluster along the perfect-prediction diagonal in the low-to-moderate PM2.5 range (0–100 $\mu g/m^3$), confirming the model is reliable for typical urban air quality levels. Deviation increases at extreme values, consistent with the RMSE of 8.41 $\mu g/m^3$ being pulled upward by outlier spike events.

Challenges and Solutions

Challenge: Plotly's `make_subplots` rendered slowly when plotting the full 134,972-row test set, causing the notebook to lag. **Solution:** Downsampled to 10,000 random points using `np.random.default_rng` before passing data to the scatter plots. The histogram uses all residuals for an accurate distribution shape since it aggregates data rather than plotting individual points.

Conclusion

Week 4 objectives were met. The three Plotly diagnostic charts provide a qualitative layer of model interpretation that goes beyond the R^2 and RMSE numbers — confirming that the pipeline's errors are largely random and non-systematic, while also transparently showing where performance degrades (extreme PM2.5 spikes). The switch from matplotlib to Plotly also satisfies the lab's professional visualization requirement by enabling interactive exploration of the results. Thank you so much!! huhu